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New potential pyrrhotite and pentlandite reference materials for sulfur and iron isotope microanalysis†

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Pyrrhotite and pentlandite are the most common Fe sulfide minerals in magmatic ore deposits and meteorites. Multiple S isotopes pairing with Fe isotopes of bulk Fe sulfides have proven to be useful tracers to constrain the formation and evolution of magmatic ore deposits. However, pyrrhotite coexists with pentlandite and other sulfides as intergrowth textures in most cases. This makes the isotope results obtained from bulk samples represent a mixed signal. High spatial resolution S and Fe isotope analytical techniques using secondary ion mass spectrometry (SIMS) and laser ablation-multiple-collector-inductively coupled plasma mass spectrometry (LA-MC-ICPMS) provide powerful tools to resolve isotopic zoning and small scale isotopic heterogeneities of Fe sulfide minerals. Matrix-matched Fe sulfide reference materials (RMs) are critical for accurate micro-beam measurements, but no natural RMs for both S–Fe isotopically homogeneous pyrrhotite and pentlandite are available. In this study, natural pyrrhotite (JC-Po) and pentlandite (JC-Pn) from the Jinchuan Ni–Cu–PGE massive sulfide ores (China) are evaluated as S and Fe isotope RMs by multiple methods. The major elements are homogeneous as verified by electron microprobe analyses (EMPA). The results of SIMS for the sulfur isotope and LA-MC-ICPMS for the iron isotope prove the homogeneity of S and Fe isotopes, respectively. From *in situ* microanalysis, the standard deviation of sulfur isotope composition is 0.27‰ (2SD, $n = 204$) for JC-Po and 0.19‰ (2SD, $n = 179$) for JC-Pn. The standard deviation of iron isotope composition is 0.33‰ (2SD, $n = 129$) for JC-Po and 0.46‰ (2SD, $n = 186$) for JC-Pn. Isotope ratio mass spectrometry (IRMS) determined that the best recommended $\delta^{34}\text{S}$ and $\delta^{33}\text{S}$ values are $0.06 \pm 0.33\text{‰}$ and $0.03 \pm 0.17\text{‰}$ (2SD, $n = 13$) for JC-Po, and $-0.09 \pm 0.15\text{‰}$ and $-0.04 \pm 0.08\text{‰}$ (2SD, $n = 13$) for JC-Pn, respectively. Based on solution MC-ICPMS analyses, the $\delta^{56}\text{Fe}$ value of $-0.34 \pm 0.07\text{‰}$ (2SD, $n = 9$) and the $\delta^{57}\text{Fe}$ value of $-0.50 \pm 0.11\text{‰}$ (2SD, $n = 9$) for JC-Po, and the $\delta^{56}\text{Fe}$ value of $1.47 \pm 0.04\text{‰}$ (2SD, $n = 4$) and the $\delta^{57}\text{Fe}$ value of $2.18 \pm 0.05\text{‰}$ (2SD, $n = 4$) for JC-Pn are provided as recommended. In summary, JC-Po pyrrhotite and JC-Pn pentlandite reference materials are suitable for *in situ* S and Fe isotope microanalysis and are expected to be used for tracking the formation processes of Fe sulfide minerals in magmatic ore deposits and meteorites soon.

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Introduction

The Fe–S system is the most important binary sulfide system because Fe sulfides are, by far, the dominant sulfide minerals in terms of geological abundance and variety of geological occurrences.¹ Constraining the S isotope signature of an individual Fe sulfide is useful in tracking the source of sulfur and sulfide precipitation processes (*e.g.*, ref. 2–4). More recently,

determining the ore-forming processes of numerous sulfide ore deposits utilized the S isotope paired with Fe isotope, including magmatic Ni–Cu–PGE deposits (*e.g.*, ref. 5–10), hydrothermal Au deposits (*e.g.*, ref. 11), Mississippi Valley-Type (MVT) Zn–Pb deposits (*e.g.*, ref. 12), and porphyry Cu deposits (*e.g.*, ref. 13). Additionally, the S and Fe isotopes of Fe sulfides from sulfide deposits on the modern seafloor and Archean shales have been investigated (*e.g.*, ref. 14–19). The original Fe isotope signal is more sensitive than the S isotope to redox reactions in a magmatic system (*e.g.*, ref. 6). The combination of Fe and S isotope bulk analyses represents a powerful new tool to constrain the evolution of magmatic ore deposits.

Pyrrhotite (Fe_{1-x}S with $0 \leq x \leq 0.125$) and pentlandite ($[(\text{Fe}, \text{Ni})_9\text{S}_8]$) are the most common Fe sulfide minerals in magmatic ore deposits and meteorites, where pyrrhotite typically coexists with pentlandite.^{20,21} Sulfur isotope determination of pyrrhotite and pentlandite has been widely used as an important tracer of

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S sources for magmatic ore deposits (e.g., ref. 22–24); especially multiple S isotope studies have yielded significant insight into the origin of magmatic sulfides (e.g., ref. 25 and references therein). A preliminary high-temperature experiment determines the equilibrium Fe isotope fractionation between pyrrhotite and silicate melt,²⁶ and a recent study indicated that pyrrhotite has potentially strong influence on the Fe isotope systematics of the magmatic system.²⁷ Almost all unaltered magmatic sulfide ores have a characteristic assemblage of pyrrhotite–pentlandite–chalcopyrite–platinum group minerals. This assemblage was formed from cooling and crystallization of a magma-derived sulfide matte, consisting predominantly of Fe, Ni, Cu, and S, fractionated to form a sequence of phases upon cooling.^{21,23} The phases, such as monosulfide solid solution, pyrrhotite, pentlandite, and intermediate solid solution, etc., are chemically distinct. The temperature changes give rise to an intergrowth texture produced by exsolution or unmixing. The variations of isotope compositions across a small mineral grain and between coexisting mineral phases can provide insights into the source, transport, and precipitation of S and Fe in the mineralization process. As individual Fe sulfide minerals differ with respect to their isotope compositions, the isotope results obtained from bulk samples represent a mixed signal. Besides, most Fe sulfide minerals in meteorites are microscopic, for instance, pyrrhotite and pentlandite in shergottites and CI chondrites.^{28,29} The above complex microscale textures make determining intra- and inter-grain isotopic heterogeneities from bulk S and Fe isotope analyses challenging. High precision *in situ* S and Fe isotope analytical techniques using LA-MC-ICPMS (e.g., ref. 30–32) and SIMS (e.g., ref. 33–40) provide powerful tools to resolve isotopic zoning and small scale isotopic heterogeneities of Fe sulfide minerals. Both LA-MC-ICPMS and SIMS analyses generate isotope ratios that are fractionated during the examinations, and this fractionation is matrix-dependent (e.g., ref. 41 and 42). A critical difficulty of *in situ* analyses is the evaluation of accuracy.⁴³ Matrix matching increases the accuracy of isotope determination, particularly for micro-beam methods. The only certified and available reference material is IRMM-014, a pure Fe metal that is almost exclusively used as the bracketing standard for Fe isotope analysis (solution and laser ablation, e.g., ref. 30 and 43–45). Natural RMs are preferred when measuring natural samples since chemical and physical properties should be as close as possible to those of the sample material for a more reliable comparison. Several well-characterized natural pyrrhotites (Anderson, Po-10, Alexo, and YP136) and pentlandites (Norilsk and VMSO) have been used for micro-beam analyses of S isotope composition.^{42,46–48} None of them could be used as an iron isotope RM for matrix-dependent Fe isotope fractionation corrections. These minerals are limited in quantity and data comparisons in different micro-beam analysis laboratories around the world. The situation is extremely unsatisfactory for mineral RMs. The requirements of RMs are both S–Fe isotopically homogeneous with a series of chemical composition and enough quantity to the scientific community. More S and Fe isotope RMs need to be developed for *in situ* microanalysis.

In this study, new natural pyrrhotite and pentlandite were reported to be used as potential RMs for the micro-beam determination of S and Fe isotopes. These RMs have large

enough quantities to be shared with other micro-beam analysis laboratories.

Sample description and preparation

The natural pyrrhotite and pentlandite samples from one hand specimen were collected from the giant Jinchuan Ni–Cu–PGE sulfide deposit on the southwestern margin of the North China Craton (38°28′32.6″N, 102°10′26.3″E). The Jinchuan deposit contains over 500 million metric tons of sulfide ores, grading 1.06 wt% Ni and 0.7 wt% Cu,^{49,50} which is hosted by a Neoproterozoic ultramafic intrusion (827 ± 8 Ma, U–Pb SHRIMP zircons in ref. 51; 831.8 ± 0.6 Ma U–Pb ID-TIMS zircons in ref. 52) within the northwest-striking Longshoushan Terrane. The Longshoushan Terrane is situated in the southwestern margin of the North China Craton and is composed of Paleoproterozoic and Mesoproterozoic metamorphic units unconformably overlain by Neoproterozoic and Paleozoic conglomerates, sandstones, and limestones.^{53,54} Two principal types of sulfide mineralization occur in this deposit: disseminated ores and net-textured (+massive) ores. The JC13-091 sulfide hand specimen (Fig. 1A) was collected from the massive ore, consisting of 70% pyrrhotite and minor amounts of pentlandite, chalcopyrite, violarite, and cubanite (Fig. 1B). Pyrrhotite and pentlandite occur as aggregates in the massive ore sample. They are highly abundant, and we handpicked *ca.* 8 g of pure pyrrhotite grains and 3 g of pure pentlandite. We crushed the pyrrhotite and pentlandite grains into fragments of around 200 μm size (Fig. 1C and D), which were placed in epoxy mounts. The mount was then polished to produce a smooth, flat sample surface with a relief of less than a few micrometers.

Analytical methods

To determine the chemical and isotope composition of potential sulfide RMs, we combined three analytical techniques.

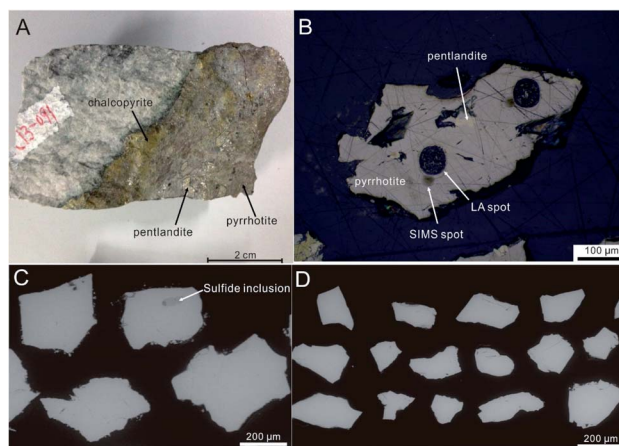


Fig. 1 Sample information for the JC13-091 pyrrhotite and pentlandite RMs, China. (A) Hand specimen of JC13-091 pyrrhotite and pentlandite. (B) Photomicrograph of JC13-091 pyrrhotite and pentlandite. (C) BSE image of the JC-Po pyrrhotite grains. (D) BSE image of the JC-Pn pentlandite grains.

Firstly, we performed wavelength dispersive spectrometry (WDS) by electron probe micro-analysis (EPMA) on multiple grains (or fragments) of each candidate reference material to ensure that all the grains are chemically homogeneous. Chemical compositions (spot measurement) were linked to backscatter electron (BSE) images to highlight potential chemical zonation, mineralogical inclusions, and fractures. Secondly, more than 100 grains of each candidate reference material were analyzed for *in situ* S and Fe isotopes by SIMS and LA-MC-ICPMS to determine the reproducibility of the grains (isotopic homogeneity). Finally, multiple grains were measured by bulk S and Fe analyses to determine the isotope compositions. Bulk S and Fe analyses were completed by isotope ratio mass spectrometry (IRMS) and solution MC-ICPMS, respectively. The respective analytical techniques are described in detail below.

EPMA

The major element chemical composition was obtained using a JEOL JXA-8100 microprobe equipped with a fully automated five X-ray WDS. The operating conditions consisted of an acceleration voltage of 15 kV, a probe current of 20 nA, and a beam diameter of 5 μm . The following sulfides were used as standards: FeS₂ (Fe, S), FeAsS (As), PbS (Pb), CuFeS₂ (Cu), ZnS (S), NiS (Ni), and CoS (Co). The detection limits for each element were Fe (126 ppm), S (126 ppm), As (478 ppm), Pb (616 ppm), Cu (111 ppm), Zn (189 ppm), Ni (96 ppm), and Co (88 ppm).

SIMS sulfur isotope measurement

In situ sulfur isotope (³²S, ³³S, and ³⁴S) analyses were performed using a Cameca IMS-1280 at the Institute of Geology and Geophysics (Beijing), Chinese Academy of Sciences. The analytical procedures and instrument parameters were similar to those described in a previous study.³⁴ Briefly, a Cs⁺ primary ion beam was accelerated at 10 kV, with an intensity of *ca.* 2.5 nA (Gaussian mode with a primary beam aperture of 400 μm to reduce aberrations). A 20 μm raster was applied during all analyses. A gold coat of ~ 30 nm thickness was applied on the sample surface, and a normal-incidence electron flood gun (NEG) was used to neutralize the possible positive charge buildup in the analysis area. The field aperture was set to 2000 μm , and the transfer-optics magnification was adjusted to give a field of view of 125 μm (FA = 8000). The energy slit width was 25 eV, and the mechanical position of the energy slit was calibrated before starting the session (5 eV gap with respect to the maximum). Sulfur isotopes were measured in multi-collector mode using three off-axis Faraday cups (L'2, L1, and H1), using an entrance slit width of 60 μm and an exit slit width of 500 μm on each of the three Faraday cup detectors positioned along the focal plane for the simultaneous measurement of ³²S, ³³S, and ³⁴S. A mass resolution power (MRP) of 2400 was employed to have flat top peaks on all the masses. A stable magnetic field was maintained using NMR (Nuclear Magnetic Resonance) regulation. The intensity of ³²S was typically 1×10^9 cps. Each analysis takes about 4.5 min, consisting of 30 s pre-sputtering, 60 s of automated centering of secondary ions, and 160 s of integrating sulfur isotope signals (40 cycles $\times$ 4 s).

The measured ³⁴S/³²S and ³³S/³²S ratios were normalized by using the Vienna-Cañon Diablo troilite (V-CDT) compositions (³⁴S/³²S = 0.044163, ³³S/³²S = 0.007877, ref. 55) as follows,

$$\delta^{34}\text{S}_{\text{measured}} \text{ in per mil} = (^{34}\text{S}/^{32}\text{S}_{\text{measured}} \div 0.044163 - 1) \times 1000$$

$$\delta^{33}\text{S}_{\text{measured}} \text{ in per mil} = (^{33}\text{S}/^{32}\text{S}_{\text{measured}} \div 0.007877 - 1) \times 1000$$

Two pyrite RMs were investigated in this study for the quality control of analyses. Sonora pyrite and Balmat pyrite were measured every eighth analysis to monitor the instrumental drift and determine the external reproducibility for each session. The Sonora pyrite was used as the primary reference reported the $\delta^{34}\text{S}$ value of 1.61‰ and $\delta^{33}\text{S}$ value of 0.83‰ (ref. 4) to calculate the instrumental mass fractionation (IMF) for the

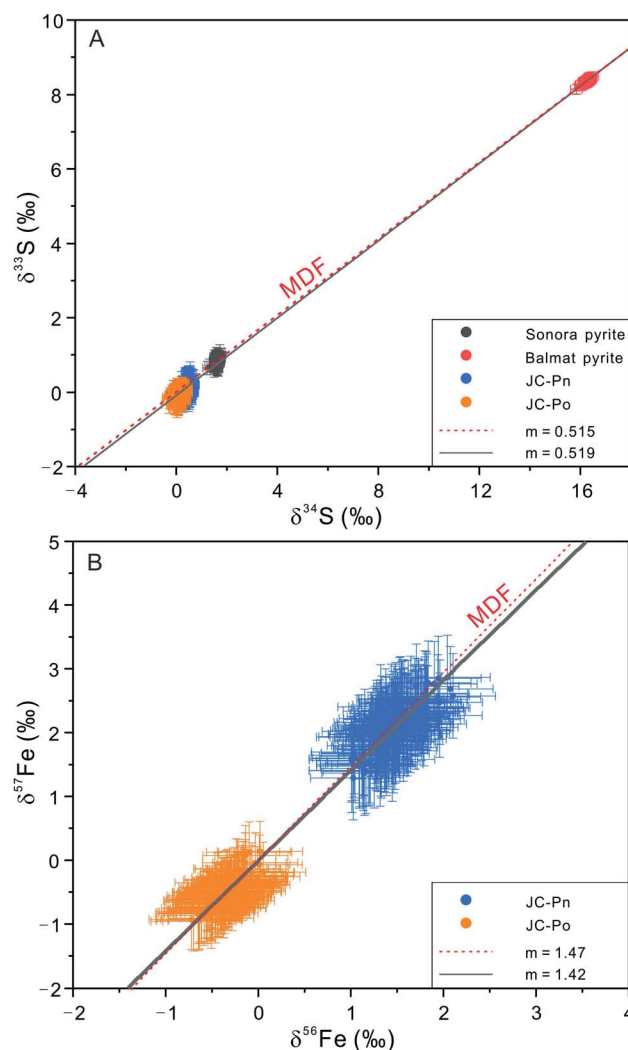


Fig. 2 Plots of (A) $\delta^{33}\text{S}$ vs. $\delta^{34}\text{S}$ and (B) $\delta^{57}\text{Fe}$ vs. $\delta^{56}\text{Fe}$ for all reference pyrites and sulfides measured in this study. Error crosses are the overall uncertainty (at 2SD) based on propagating the individual analysis repeatability with the reproducibility for the relevant analytical session. All the data are plotted on a near-ideal trend (red dash line) for mass-dependent isotope fractionation (gray solid line).

Balmat pyrite. Also, the $\delta^x\text{S}_{\text{measured}}$ values were normalized with the average value of IRMS measurements following the equation below,

$$\delta^x\text{S}_{\text{final}} = \delta^x\text{S}_{\text{measured}} + (\text{mean of } \delta^x\text{S}_{\text{dual-inlet IRMS}} - \text{mean of } \delta^x\text{S}_{\text{measured}})$$

where $x = 34, 33$.

In this study, the Balmat pyrite measurements yielded a mean $\delta^{34}\text{S}$ value of $16.25 \pm 0.23\text{‰}$ ($n = 80$), which within error is consistent with previously reported values.³⁴ For Sonora and Balmat pyrite RMs, mass-dependent fractionation (MDF) was reported.^{4,33,46} All *in situ* S isotope data from this study are in plots of $\delta^{33}\text{S}$ vs. $\delta^{34}\text{S}$ (Fig. 2A), with the average and standard deviations of the measurements. The $\delta^{33}\text{S}$ and $\delta^{34}\text{S}$ values reveal a slope of 0.513 ($n = 605$, $R^2 = 0.99$), which is in good agreement with the MDF ratio of 0.515.⁵⁶ The *in situ* S isotope measurements of pyrite RMs prove the data obtained by SIMS are reliable at IGGCAS.

IRMS sulfur isotope measurement

A total of 26 samples were selected randomly for sulfur isotope analysis. 13 measurements for each sample were performed in four different laboratories. Pure pyrrhotite and pentlandite grains, 0.5 to 2 mm in diameter, were handpicked under a binocular microscope.

Two samples were analyzed by dual-inlet portion IRMS at the stable isotope laboratory at the MNR (Ministry of Natural Resources of the People's Republic of China) Key Laboratory of Metallogeny and Mineral Assessment, Institute of Mineral Resources, Chinese Academy of Geological Sciences (CAGS). Dual-inlet portion IRMS analyzed two samples at the stable isotope laboratory at the Analytical Laboratory of Beijing Research Institute of Uranium Geology. The dual-inlet portion of the analysis is a computer-controlled routine (Thermo Fisher Scientific Inc.). An 8 mg pyrrhotite/pentlandite sample was combined with Cu_2O and finely ground in an agate mortar. The samples were combusted under vacuum at 900°C , producing SO_2 gas. The SO_2 was cryogenically purified and analyzed mass spectrometrically on a MAT 251 mass spectrometer using the standard dual-inlet protocol at the above two laboratories.

The remaining 22 samples were analyzed with an elemental analyzer coupled to a mass spectrometer (EA-IRMS). Approximately 0.1 mg of sulfide powder was combined with V_2O_5 and combusted in an elemental analyzer. Sulfur isotope ratios were determined with a MAT 251 mass spectrometer. Eight samples were analyzed at the MNR Key Laboratory of Metallogeny and Mineral Assessment, Institute of Mineral Resources, CAGS. Four samples were analyzed at the Analytical Laboratory of Beijing Research Institute of Uranium Geology. Six samples were analyzed at the State Key Laboratory of Environmental Geochemistry, Institute of Geochemistry, Chinese Academy of Sciences. Four samples were analyzed at the Geology, Geophysics, and Geochemistry Science Center, U.S. Geological Survey, Denver. The sulfur isotope composition is expressed in standard δ notation as per mil (‰) deviations from V-CDT.

LA-MC-ICP-MS iron isotope measurement

The homogeneity test of iron isotopes from pyrrhotite and pentlandite was conducted at the Beijing Createch Testing Technology Co., Ltd. The Fe isotope measurements were performed by using a high mass resolution MC-ICPMS (Thermo-Finnigan Neptune Plus) connected to a RESolution-193 laser ablation system. The analytical conditions involved an ablation pit of $35\text{ }\mu\text{m}$ diameter, an ablation time of 30 s, a repetition rate of 3 Hz, and a laser beam energy density of 10 J cm^{-2} . The LA-MC-ICPMS Fe isotope measurements were performed at high mass resolution ($M/\Delta M \approx 9000$, 5–95% peak side width definition) to resolve the molecular interference of argon nitrides and argon oxides on Fe isotopes.⁵⁷ A detector array of Faraday cups is set to acquire the ion beams of ^{53}Cr , ^{54}Fe , ^{56}Fe , ^{57}Fe , ^{58}Fe , and ^{60}Ni . Cr and Ni are monitored and used for the correction of the isobaric interference on ^{54}Fe and ^{58}Fe . The BCR-2G standard and ZX pyrite in-house standard were measured before every 10 sample analyses for drift monitoring.

The measured $^{56}\text{Fe}/^{54}\text{Fe}$ and $^{57}\text{Fe}/^{54}\text{Fe}$ ratios were converted to $\delta^{56}\text{Fe}$ and $\delta^{57}\text{Fe}$ values through normalizing by using the IRMM-014 compositions ($^{56}\text{Fe}/^{54}\text{Fe} = 15.6979$, $^{57}\text{Fe}/^{54}\text{Fe} = 0.3625$, ref. 58) as follows,

$$\delta^{56}\text{Fe}_{\text{measured}} \text{ in per mil} = (^{56}\text{Fe}/^{54}\text{Fe}_{\text{measured}} \div 15.6979 - 1) \times 1000$$

$$\delta^{57}\text{Fe}_{\text{measured}} \text{ in per mil} = (^{57}\text{Fe}/^{54}\text{Fe}_{\text{measured}} \div 0.3625 - 1) \times 1000$$

The $\delta^i\text{Fe}_{\text{measured}}$ values were normalized with the average value of solution MC-ICPMS measurements following the equation below,

$$\delta^i\text{Fe}_{\text{final}} = \delta^i\text{Fe}_{\text{measured}} + (\text{mean of } \delta^i\text{Fe}_{\text{solution MC-ICPMS}} - \text{mean of } \delta^i\text{Fe}_{\text{measured}})$$

where $i = 56, 57$.

In addition, the BCR-2G reference was investigated in this study to monitor the stability of isotopic ratios. In order to evaluate the data quality with respect to potential interference from molecular species and isobaric overlaps of Cr, samples from each analytical session were plotted on a $\delta^{57}\text{Fe}$ vs. $\delta^{56}\text{Fe}$ diagram (Fig. 2B). The $\delta^{57}\text{Fe}$ and $\delta^{56}\text{Fe}$ values reveal a slope of 1.42 ($n = 317$, $R^2 = 0.99$), which is in good agreement with the mass-dependent fractionation ratio of 1.47 that is based on the natural abundances of Fe isotopes.⁵⁹ This suggests the insignificant contribution of isobaric or molecular overlapping on ^{54}Fe .³⁰

Solution MC-ICP-MS iron isotope measurement

The chemical separation of Fe isotopic compositions was undertaken at the MNR Key Laboratory of Metallogeny and Mineral Assessment, Institute of Mineral Resources, CAGS, following the procedure described by Hou *et al.*⁶⁰ The samples of pyrrhotite and pentlandite were dissolved in aqua regia. After drying down the dissolved procedure, the samples were dissolved in nitric acid to completely oxidize iron to Fe^{3+} . The Fe

and other elements were separated effectively by the method of ion-exchange chromatography.

The determination of iron isotope compositions was undertaken at the MNR Key Laboratory of Isotopic Geology, Institute of Geology, CAGS. Iron isotope ratios were determined in high mass resolution mode on a Nu Plasma HR MC-ICP-MS using the standard-sample bracketing (SSB) approach. Samples were introduced into the mass spectrometer in 0.1 mol L⁻¹ HNO₃ using a DSN 100 desolvating nebulizer with Fe concentrations at 5 ppm. The Fe isotope results are expressed as deviations of an iron isotope ratio of a sample from that of the reference material IRMM-014 in per mil: $\delta^i\text{Fe} (\text{‰}) = [({}^i\text{Fe}/{}^{54}\text{Fe})_{\text{sample}}/({}^i\text{Fe}/{}^{54}\text{Fe})_{\text{standard}} - 1] \times 1000$, and $i = 56, 57$. The long-term external reproducibility in high-resolution mode is better than 0.05‰ for $\delta^{56}\text{Fe}$ at 2SD.⁶¹

Results and discussion

Chemical compositions of reference materials

The iron and sulfur compositions of JC-Po pyrrhotite and JC-Pn pentlandite are presented in Fig. 3. The chemical compositions of JC-Po are fairly homogeneous, as the concentration of Fe ranges from 58.99 wt% to 61.38 wt%, and the concentration of S ranges from 38.25 wt% to 39.87 wt%. Other elements ($n = 63$) with lower concentrations are As 0.05 wt%, Pb 0.03 wt%, Cu 0.01 wt%, Zn 0.01 wt%, Ni 0.43 wt%, and Co 0.12 wt%, respectively. The chemical compositions of JC-Pn are also homogeneous, as the concentration of Fe ranges from 29.40 wt% to 30.97 wt%, the concentration of Ni ranges from 35.17 to 36.05 wt%, and the concentration of S ranges from 32.14 wt% to 32.85 wt%. Other elements ($n = 20$) with lower concentrations are As 0.03 wt%, Pb 0.02 wt%, Cu 0.02 wt%, and Co 1.14 wt%. The detailed data of individual spots are listed in ESI Table S1.† The BSE image (Fig. 1C) of JC-Po shows no internal growth or zonation. A few sulfide mineral inclusions

may present in some fragments. Thus the mineral inclusions need to be avoided during micro-beam analyses. The BSE image (Fig. 1D) shows that most JC-Pn pentlandite fragments are homogeneous without mineral inclusions.

Homogeneity of reference materials

Homogeneity is of first-order importance for reference materials, and it was achieved here through the application of national regulations for reference material preparation (ISO Guide 35 2017).⁶² Fifteen grains of each pyrrhotite and pentlandite were randomly selected to test the homogeneity of isotope ratios from JC-Po pyrrhotite and JC-Pn pentlandite, respectively. Two subsamples were taken from each grain (Fig. 1B), with each being treated as an independent sample. The results of isotope analyses of the replicates are expressed as X_{i1} and X_{i2} in ESI Tables S2 and S3,† and the average of duplicate pairs as $\bar{X}$. Homogeneity testing was conducted under strictly the same conditions. All tests were conducted in a single laboratory, using the same analytical method, by one analyst, and with all samples tested in the same session.

Inhomogeneity between grains was tested by using single-factor ANOVA statistics. The experimental F ratio is the ratio of the among-grain variance (S_{among}^2) to the within-grain variance (S_{within}^2):

$$F = \frac{S_{\text{among}}^2}{S_{\text{within}}^2} \quad (1)$$

where S_{among}^2 is the ratio of among-grain sums of squares (SS_{among}) to the associated among-grain degrees of freedom (ν_{among}), and S_{within}^2 is the ratio of within-grain sums of squares (SS_{within}) to the associated within-grain degrees of freedom (ν_{within}):

$$S_{\text{among}}^2 = \frac{SS_{\text{among}}}{\nu_{\text{among}}} \quad (2)$$

$$S_{\text{within}}^2 = \frac{SS_{\text{within}}}{\nu_{\text{within}}} \quad (3)$$

where ν_{among} and ν_{within} depend on the number of units from which samples are taken (m) and the number of replicate measurements for each grain (n), with ν_{among} and ν_{within} computed as follows:

$$\nu_{\text{among}} = m - 1 \quad (4)$$

$$\nu_{\text{within}} = m(n - 1) \quad (5)$$

The results shown in the ANOVA tables were computed (Table 1). F -Testing demonstrates that the samples have very good homogeneity [$F < F_{\text{critical}}(\nu_{\text{among}}, \nu_{\text{within}})$; critical value of F for $\alpha = 5\%$].^{63,64} Differences in isotopic ratios (Table 1) were caused mainly by the repeatability of the method, rather than the heterogeneity of the reference materials.

S isotope compositions of the reference materials

Sulfur isotope compositions determined at the four different laboratories were consistent. All sulfides were supplied as single

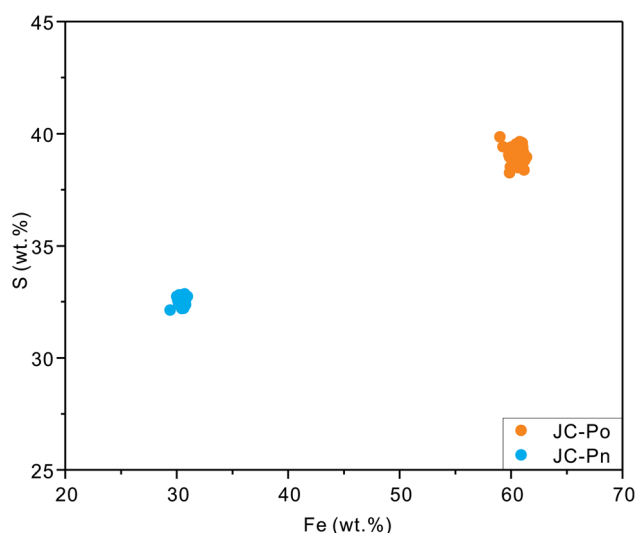


Fig. 3 Electron microprobe analytical results of JC-Po pyrrhotite and JC-Pn pentlandite from the Jinchuan deposit. Diagram illustrating the direct correlation between Fe and S.

Table 1 ANOVA table for homogeneity testing of S and Fe isotope ratios of JC-Po pyrrhotite and JC-Pn pentlandite

Sample	Isotopic ratio	$\bar{X}$	SS _{among}	SS _{within}	F	F _{0.05(14,15)} ^a
JC-Po	³⁴ S/ ³² S	0.0441680	$1.8139605 \times 10^{-10}$	$4.0723005 \times 10^{-10}$	0.48	2.42
	³³ S/ ³² S	0.0078763	$7.4492277 \times 10^{-12}$	$1.0447738 \times 10^{-11}$	0.76	
	⁵⁶ Fe/ ⁵⁴ Fe	16.7786450	4.0897834×10^{-5}	2.9132698×10^{-5}	1.50	
	⁵⁷ Fe/ ⁵⁴ Fe	0.3998597	6.6523953×10^{-8}	4.4104760×10^{-8}	1.62	
JC-Pn	³⁴ S/ ³² S	0.0441835	$3.2271119 \times 10^{-10}$	$1.6206435 \times 10^{-10}$	2.13	2.42
	³³ S/ ³² S	0.0078780	$2.9067739 \times 10^{-12}$	$1.4138295 \times 10^{-12}$	2.20	
	⁵⁶ Fe/ ⁵⁴ Fe	16.8267616	1.3285750×10^{-4}	6.6992162×10^{-5}	2.12	
	⁵⁷ Fe/ ⁵⁴ Fe	0.4014181	1.0092676×10^{-7}	1.3995443×10^{-7}	0.77	

^a $n = 2$; $m = 3$; $\nu_{\text{among}} = 14$; $\nu_{\text{within}} = 15$.

samples. Each sample was requested to measure at least one duplicate in each laboratory. The intercomparison results are given in Table 2, with data presented as means $\pm$ 2SD (Fig. 4A and C).

The sulfur isotope compositions of RMs are shown in Table 2. The sulfur isotope RMs GBW04414 (-0.07‰), GBW04415 (22.15‰), and IAEA-S-3 (-32.49‰) were measured in the MNR Key Laboratory of Metallogeny and Mineral Assessment, Institute of Mineral Resources, CAGS. The sulfur isotope RMs IAEA-S-1 (-0.3‰), IAEA-S-2 (22.62‰), and IAEA-S-3 (-32.49‰) were measured in the other three laboratories. All the results of RMs were consistent with the recommended values to prove that accurate S isotope measurements were able to be obtained in these four laboratories.

Eleven measurements of sulfur isotopes from each sample were conducted by EA-IRMS. The results are shown in Table 2.

The JC-Po values of $\delta^{34}\text{S}$ ranged from -0.09‰ to 0.40‰ (Fig. 4A). The JC-Pn values of $\delta^{34}\text{S}$ ranged from -0.20‰ to 0.10‰ (Fig. 4C). Two measurements by dual-inlet IRMS for each sample yielded consistent values with the results determined by EA-IRMS. All thirteen IRMS measurements gave a mean $\delta^{34}\text{S}$ value of $0.06 \pm 0.33\text{‰}$ (2SD, $n = 13$) for JC-Po pyrrhotite. Thirteen IRMS analyses for JC-Pn pentlandite gave a mean $\delta^{34}\text{S}$ value of $-0.09 \pm 0.15\text{‰}$ (2SD, $n = 13$). These grand mean $\delta^{34}\text{S}$ values are interpreted as the best recommended values for JC-Po pyrrhotite and JC-Pn pentlandite, respectively.

A total of 204 sulfur isotope ratio analyses were randomly performed on the fragments of JC-Po pyrrhotite. The data were normalized to a $\delta^{34}\text{S}$ value of $0.06 \pm 0.33\text{‰}$ (2SD, $n = 13$) determined by IRMS (Fig. 4B). The measurement repeatability (2SD, $n = 204$) for $\delta^{34}\text{S}$ and $\delta^{33}\text{S}$ was 0.27‰ and 0.24‰ , respectively. A total of 179 sulfur isotope ratio analyses were

Table 2 IRMS Sulfur Isotope Results for JC-Po pyrrhotite and JC-Pn pentlandite^a

Lab.	Replicates	JC-Po	JC-Pn	S1/GBW04414 ^b	S2/GBW04415 ^b	S3
		$\delta^{34}\text{S}$ (‰)	$\delta^{34}\text{S}$ (‰)	$\delta^{34}\text{S}$ (‰)	$\delta^{34}\text{S}$ (‰)	$\delta^{34}\text{S}$ (‰)
1	1	0.00 ^c	0.00 ^c	-0.07	21.98	-32.30
	2	0.30	-0.10	-0.03	22.09	-32.50
	3	0.12	-0.16	—	22.28	-32.74
	4	-0.09	-0.11	—	—	—
	5	-0.08	-0.11	—	—	—
2	Mean	0.05 ± 0.33 (2SD)	-0.10 ± 0.12 (2SD)	-0.05 ± 0.06 (2SD)	22.12 ± 0.30 (2SD)	-32.51 ± 0.44 (2SD)
	1	-0.08 ^c	-0.10 ^c	-0.40	22.47	-32.31
	2	0.10	-0.10	-0.09	22.56	-32.22
	3	0.10	0.10	-0.24	22.45	—
	Mean	0.04 ± 0.21 (2SD)	-0.03 ± 0.23 (2SD)	-0.24 ± 0.31 (2SD)	22.49 ± 0.12 (2SD)	-32.51 ± 0.13 (2SD)
3	1	-0.09	-0.15	-0.23	22.57	-32.35
	2	-0.05	-0.04	-0.13	22.49	-32.29
	3	-0.09	-0.05	-0.12	22.47	-32.43
	Mean	-0.08 ± 0.05 (2SD)	-0.08 ± 0.12 (2SD)	-0.16 ± 0.12 (2SD)	22.51 ± 0.11 (2SD)	-32.36 ± 0.14 (2SD)
	4	0.40	-0.20	-0.26	22.42	-32.49
4	2	0.20	-0.10	-0.27	22.51	-32.78
	Mean	-0.30 ± 0.05 (2SD)	-0.15 ± 0.12 (2SD)	-0.27 ± 0.01 (2SD)	22.47 ± 0.13 (2SD)	-32.64 ± 0.41 (2SD)
	Grand mean	0.06 ± 0.33 (2SD)	-0.09 ± 0.15 (2SD)	—	—	—

^a Lab. 1: Stable Isotope Laboratory at the MNR Key Laboratory of Metallogeny and Mineral Assessment, Institute of Mineral Resources, CAGS; Lab. 2: Stable Isotope Laboratory at the Analytical Laboratory of Beijing Research Institute of Uranium Geology; Lab. 3: Stable Isotope Laboratory at the State Key Laboratory of Environmental Geochemistry, Institute of Geochemistry, CAS; Lab. 4: Stable Isotope Laboratory at the Geology, Geophysics, and Geochemistry Science Center, USGS. ^b The results of IRMS from Lab. 1 were monitored through multiple analyses of the GBW04414, GBW04415, and IAEA-S-3 RMs. The results of IRMS from other labs were monitored through multiple analyses of IAEA-S-1, IAEA-S-2, and IAEA-S-3 RMs. ^c The measurements used the dual-inlet IRMS at Lab. 1 and 2. Others used the EA-IRMS at the four laboratories.

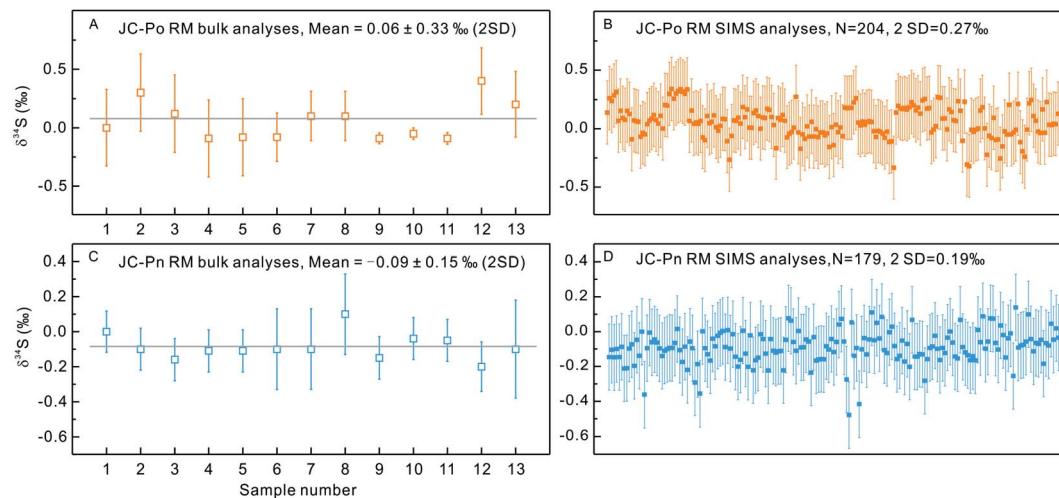


Fig. 4 Sulfur isotope compositions of JC-Po pyrrhotite and JC-Pn pentlandite. (A) The $\delta^{34}\text{S}$ values of JC-Po pyrrhotite ($n = 13$) determined by IRMS; (B) homogeneity test of sulfur isotopes from JC-Po pyrrhotite by SIMS; (C) the $\delta^{34}\text{S}$ values of JC-Pn pentlandite ($n = 13$) determined by IRMS; (D) homogeneity test of sulfur isotopes from JC-Pn pentlandite by SIMS. All spot analyses were performed through SIMS in an overnight run (~ 17 hours) without drift correction, which shows limited variation and demonstrated that the grains are homogeneous in terms of sulfur isotope composition. Note the SIMS data were normalized to the $\delta^{34}\text{S}$ value determined by IRMS.

randomly performed on the fragments of JC-Pn pentlandite. The data were normalized to a $\delta^{34}\text{S}$ value of $-0.09 \pm 0.15\text{‰}$ (2SD, $n = 13$) determined by IRMS (Fig. 4D). The measurement repeatability (2SD, $n = 179$) for $\delta^{34}\text{S}$ and $\delta^{33}\text{S}$ was 0.19‰ and 0.29‰ , respectively. The data show limited variations with excellent repeatability (Fig. 4B and D), and the raw data are given in ESI Table S4.†

The distinct change in the MIF S isotope signature of sedimentary sulfide and sulfate minerals occurred during the first great oxidation event (GOE) at about 2.4–2.3 Ga. Prior to this first GOE, these minerals record distinct signatures of mass-independently fractionated sulfur isotopes, whereas afterwards this signature disappears completely.⁶⁵ JC-Po pyrrhotite and JC-Pn pentlandite from the Jinchuan deposit formed at *ca.* 830 Ma (after GOE). As any mass-independent fractionation during *in situ* S isotope measurements was indistinguishable (Fig. 2A), $\delta^{33}\text{S}$ values can be simply calculated as $0.515\delta^{34}\text{S}$ values based on the MDF law.³³ The $\delta^{33}\text{S}$ value of JC-Po pyrrhotite was calculated to be $0.03 \pm 0.17\text{‰}$ (2SD, $n = 13$), and the $\delta^{33}\text{S}$ value of JC-Pn pentlandite was calculated to be $-0.04 \pm 0.08\text{‰}$ (2SD, $n = 13$).

Fe isotope compositions of the reference materials

Iron isotope compositions were determined by one laboratory at CAGS. Nine samples for JC-Po pyrrhotite and four samples for JC-Pn pentlandite were provided as single samples. The results are given in Table 3, and Fig. 5, with data presented as means $\pm$ 2SD. The JC-Po values of $\delta^{56}\text{Fe}$ ranged from -0.39‰ to -0.29‰ , with mean $\delta^{56}\text{Fe}$ values of $-0.34 \pm 0.07\text{‰}$ (2SD, $n = 9$, Fig. 5A). The JC-Pn values of $\delta^{56}\text{Fe}$ ranged from 1.44‰ to 1.49‰ , with mean $\delta^{56}\text{Fe}$ values of $1.47 \pm 0.04\text{‰}$ (2SD, $n = 9$, Fig. 5C). The mean $\delta^{57}\text{Fe}$ values of JC-Po and JC-Pn were $-0.50 \pm 0.11\text{‰}$ (2SD, $n = 9$) and $2.18 \pm 0.05\text{‰}$ (2SD, $n = 4$), respectively. All the grand

mean values are interpreted as the best recommended Fe isotope values for JC-Po pyrrhotite and JC-Pn pentlandite.

A total of 129 *in situ* iron isotope ratio analyses for JC-Po pyrrhotite were conducted. The data were normalized to the $\delta^{56}\text{Fe}$ value of $-0.34 \pm 0.07\text{‰}$ (2SD, $n = 9$) by solution MC-ICPMS (Table 3, Fig. 5B). The measurement repeatability (2SD, $n = 129$) for $\delta^{56}\text{Fe}$ and $\delta^{57}\text{Fe}$ was 0.33‰ and 0.47‰ , respectively. A total of 186 *in situ* iron isotope ratio analyses for JC-Pn pentlandite were conducted. The data were normalized to the $\delta^{56}\text{Fe}$ value of $1.47 \pm 0.04\text{‰}$ (2SD, $n = 4$) by solution MC-ICPMS (Table 3, Fig. 5D). The measurement repeatability (2SD, $n = 186$) for $\delta^{56}\text{Fe}$ and $\delta^{57}\text{Fe}$ was 0.46‰ and 0.66‰ , respectively. All the spot analysis data show limited variations

Table 3 Solution MC-ICPMS iron isotope results for JC-Po pyrrhotite and JC-Pn pentlandite

Sample	$\delta^{56}\text{Fe}$ (‰)	2SD (‰)	$\delta^{57}\text{Fe}$ (‰)	2SD (‰)
JC-Po	-0.31	0.02	-0.43	0.10
	-0.37	0.04	-0.54	0.05
	-0.34	0.05	-0.48	0.06
	-0.38	0.03	-0.56	0.11
	-0.31	0.06	-0.47	0.11
	-0.39	0.04	-0.58	0.13
	-0.36	0.08	-0.55	0.12
	-0.29	0.07	-0.45	0.10
	-0.31	0.00	-0.47	0.12
Mean	$-0.34 \pm 0.07\text{‰}$ (2SD)		$-0.50 \pm 0.11\text{‰}$ (2SD)	
JC-Pn	1.48	0.06	2.18	0.08
	1.44	0.05	2.15	0.08
	1.47	0.06	2.21	0.11
	1.49	0.05	2.19	0.12
Mean	$1.47 \pm 0.04\text{‰}$ (2SD)		$2.18 \pm 0.05\text{‰}$ (2SD)	

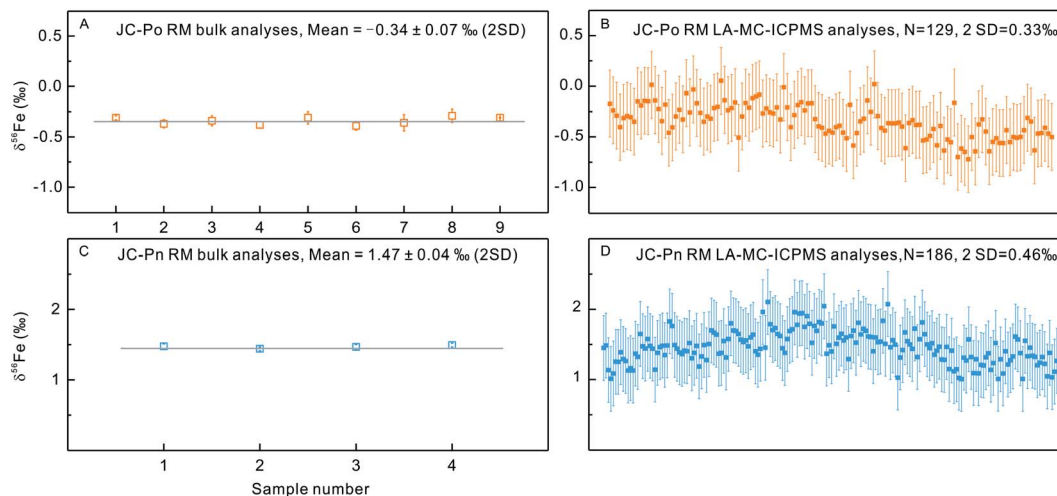


Fig. 5 Iron isotope compositions of JC-Po pyrrhotite and JC-Pn pentlandite. (A) The $\delta^{56}\text{Fe}$ values of JC-Po pyrrhotite ($n = 9$) determined by solution MC-ICPMS; (B) homogeneity test of iron isotopes from JC-Po pyrrhotite by LA-MC-ICPMS; (C) the $\delta^{56}\text{Fe}$ values of JC-Pn pentlandite ($n = 4$) determined by solution MC-ICPMS; (D) homogeneity test of iron isotopes from JC-Pn pentlandite by LA-MC-ICPMS. Spot analyses of JC-Po pyrrhotite (B) and JC-Pn pentlandite (D) through LA-MC-ICPMS in a continuous run (~ 7 hours) without drift correction, which shows limited variation and demonstrated that the grains are homogeneous in terms of iron isotope compositions. Note the LA-MC-ICPMS data were normalized to the $\delta^{56}\text{Fe}$ value determined by solution MC-ICPMS.

with excellent repeatability (Fig. 5B and D), and the raw data are given in ESI Table S5.†

Conclusions

This study has demonstrated that Jinchuan pyrrhotite (JC-Po) and pentlandite (JC-Pn) from ore hand specimen JC13-091 are homogeneous in S and Fe isotope compositions. These two iron sulfide minerals are suitable as RMs for S and Fe isotope microanalysis using SIMS and LA-MC-ICPMS, respectively. For JC-Po pyrrhotite, the best recommended $\delta^{34}\text{S}$, $\delta^{33}\text{S}$, $\delta^{56}\text{Fe}$, and $\delta^{57}\text{Fe}$ values are $0.06 \pm 0.33\text{‰}$ (2SD, $n = 13$), $0.03 \pm 0.17\text{‰}$ (2SD, $n = 13$), $-0.34 \pm 0.07\text{‰}$ (2SD, $n = 9$), and $-0.50 \pm 0.11\text{‰}$ (2SD, $n = 9$), respectively. For JC-Pn pentlandite, the best recommended $\delta^{34}\text{S}$, $\delta^{33}\text{S}$, $\delta^{56}\text{Fe}$, and $\delta^{57}\text{Fe}$ values are $-0.09 \pm 0.15\text{‰}$ (2SD, $n = 13$), $-0.04 \pm 0.08\text{‰}$ (2SD, $n = 13$), $1.47 \pm 0.04\text{‰}$ (2SD, $n = 4$), and $2.18 \pm 0.05\text{‰}$ (2SD, $n = 4$), respectively.

There are currently 8 g of JC-Po and 3 g of JC-Pn grains available for distribution. They are stored in the SIMS laboratory at the Institute of Geology and Geophysics, Chinese Academy of Sciences in Beijing, and are available on request from the corresponding authors of this article.

Author contributions

Lei Chen and Xian-Hua Li conceived the presented idea. Yu Liu and Lei Chen carried out the experiments. Yang Li and Qiu-Li Li contributed to the interpretation of the results. Lei Chen took the lead in writing the manuscript. Xian-Hua Li encouraged Lei Chen to investigate the reference materials for microanalysis and supervised the project. All the authors provided critical feedback and helped shape the research, analysis, and manuscript.

Conflicts of interest

The authors declare that they have no conflict of interest.

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